

A PARAMETER OPTIMISATION IN SAWIN'S EXPLICIT UNIT-DISTANCE LOWER BOUND

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ABSTRACT. Sawin recently made explicit the exponent in the OpenAI construction of planar point sets with superlinear many unit distances, obtaining an exponent 1.014114 by a completely explicit refinement of the number-theoretic argument. This note records a computer-assisted optimisation of the finite parameters in Sawin's proof, with no new theoretical input. A prefix choice of T already gives exponent 1.031722..., and a targeted non-prefix replacement

$$T = (\text{first 81 odd primes}) \setminus \{197, 337\} \cup \{433, 601\}$$

improves the verified exponent to 1.031750838.... Thus the same proof gives the rounded lower-bound exponent 1.03175.

1. INTRODUCTION

Let $N(U)$ denote the number of ordered unit-distance pairs in a finite set $U \subset \mathbb{R}^2$; changing to unordered pairs only changes absolute constants. The classical unit-distance problem asks how large $N(U)$ can be in terms of $n = \#U$. Erdős gave the long-standing lattice lower bound of shape $n^{1+c/\log \log n}$ and conjectured, in one form, that $n^{1+o(1)}$ should be the correct order of magnitude [6, 7, 8]. The best general upper bound remains $O(n^{4/3})$, due to Spencer, Szemerédi and Trotter [19].

The recent OpenAI paper [3] is ground-breaking: it gives the first construction with $N(U) \geq n^{1+\delta}$ for infinitely many n and some fixed $\delta > 0$, thereby disproving the traditional $n^{1+o(1)}$ expectation. Its method is also striking because it passes through unramified towers of number fields, Golod–Shafarevich theory, and high-dimensional lattice projections. The human-digested remarks of Alon–Bloom–Gowers–Litt–Sawin–Shankar–Tsimmerman–Wang–Wood [4] clarify the proof and its provenance, while Bubeck et al. [5] place this and related developments in the broader context of AI-assisted research.

Sawin's note [2] makes the OpenAI construction explicit and sharper. His Theorem 1 gives, for arbitrarily large n , sets $U \subset \mathbb{R}^2$ with

$$N(U) \geq n^{1.014114}/C$$

for an absolute constant C . The proof reduces the last numerical step to the choice of finite data T , $S_{\mathbb{Q}}$, k and R in Lemma 12 and Proposition 10 of [2]. Sawin explicitly notes that these parameters were not optimised very carefully.

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This note records the result of optimising precisely those parameters, first in the prefix family for T and then by a small splitting-pattern modification of T .

The verification code, the JSON certificates, and a Jupyter notebook interface are in the repository

<https://github.com/tomaszkania/unit-distance-gs-optimiser>.

The repository is cited as [1]. The code checks the primality, congruence, splitting and Golod–Shafarevich conditions exactly, and evaluates the logarithmic expression numerically.

2. THE IMPROVED NUMERICAL CERTIFICATE

We use Sawin’s notation. Let

$$T_0 = \{3, 5, 7, \dots, 421\}$$

be the first 81 odd primes, and put

$$T = T_0 \setminus \{197, 337\} \cup \{433, 601\}.$$

Then $\#T = 81$, and exactly 41 elements of T are congruent to 3 modulo 4, so the parity condition in Lemma 12 of [2] is satisfied. Put

$$Q = \mathbb{Q} \left(\sqrt{\prod_{q \in T} q} \right).$$

Sawin’s Golod–Shafarevich condition (9) is

$$\#T + \#S_{\mathbb{Q}} + \#\{p \in S_{\mathbb{Q}} : p \text{ splits in } Q\} + 1 \leq \frac{(\#T - 1)^2}{4}.$$

For $\#T = 81$ this gives the budget

$$\#S_{\mathbb{Q}} + \#\{p \in S_{\mathbb{Q}} : p \text{ splits in } Q\} \leq 1518.$$

The certificate in the repository selects $\#S_{\mathbb{Q}} = 1278$ primes, of which 240 split in Q . Thus

$$\#S_{\mathbb{Q}} + \#\{p \in S_{\mathbb{Q}} : p \text{ splits in } Q\} = 1278 + 240 = 1518,$$

so condition (9) is met with equality.

For each selected prime, the certificate also gives a positive integer $k(p)$. The largest selected prime is 16007. The verifier recomputes, rather than trusts, the ramification index

$$e(p) = \begin{cases} 2, & p = 2 \text{ or } p \in T, \\ 1, & \text{otherwise,} \end{cases}$$

the local admissibility condition of Lemma 12, and whether p splits in Q .

With

$$R = 32.49523163513887\dots,$$

Sawin's equation (11) gives $\delta = A/B$, where

$$A = \log(1 - 1/R) + \frac{1}{2} \log(2\pi/e) + \sum_{p \in S_{\mathbb{Q}}} \frac{1}{4e(p)} \log(k(p) + 1) \\ - \frac{1}{8} \log \left(4 \prod_{q \in T} q \right) - \frac{1}{2} \log \log \sqrt{4 \prod_{q \in T} q}$$

and

$$B = \log \left(2R \prod_{p \in S_{\mathbb{Q}}} p^{k(p)/(2e(p))} + 1 \right).$$

The certificate evaluates the numerator and denominator as

$$170.951353249675\dots, \quad 5384.152468939053\dots,$$

respectively, and hence

$$\delta = 0.0317508380819239\dots$$

The displayed denominator includes the +1 inside the logarithm; at this size the correction is far beyond the displayed precision, but it is included in the verifier.

Theorem 1. *For arbitrarily large n , there is a set $U \subset \mathbb{R}^2$ with $\#U = n$ and*

$$N(U) \geq n^{1.03175}/C$$

for an absolute constant C .

Proof. The exact congruence and splitting checks in the certificate verify the hypotheses of Lemma 12 of [2]. The equality in the Golod–Shafarevich budget above verifies condition (9). Therefore Lemma 12 supplies the family of totally real fields F and CM extensions $K = F(i)$ required by Sawin's Proposition 10.

Substituting the certified T , $S_{\mathbb{Q}}$, k and R into Sawin's equation (11) gives $\delta = 0.0317508380819239\dots$. Proposition 10 therefore gives

$$N(U) \geq \frac{(\#U)^{1+\delta}}{8\lambda^2}, \quad \lambda = \sqrt{4 \prod_{q \in T} q},$$

for arbitrarily large $\#U$. Since $\delta > 0.03175$ and λ is fixed, the asserted bound follows after absorbing the fixed factor into C . $\square$

3. HOW THE OPTIMISER CHOOSES THE PARAMETERS

The objective is a ratio. For a trial value of δ , the code maximises

$$\text{numerator} - \delta \text{ denominator}.$$

Ignoring only the negligible $+1$ inside the final logarithm during the search, the optimal value of R for the transformed objective is

$$R = 1 + \frac{1}{\delta}.$$

For a candidate prime p , the best exponent is found by maximising

$$\frac{\log(k+1)}{4e(p)} - \delta \frac{k \log p}{2e(p)}$$

over positive integers k . Once these local scores are known, selecting $S_{\mathbb{Q}}$ is a two-weight knapsack problem: a non-split prime costs 1 unit of the Golod–Shafarevich budget, and a split prime costs 2 units. Since the weights are only 1 and 2, sorting the two classes separately and comparing prefix sums solves the selection problem efficiently.

This optimiser is then wrapped in a bisection search for the value at which the transformed objective changes sign. The final result is not accepted from the optimiser alone: the stored JSON certificate is checked independently by the verifier. The notebook API exposes the same checks through functions such as `load_certificate_summary` and `search_t_summary`.

4. THE SPLITTING-PATTERN SWAP

For the prefix set T_0 , the same optimiser gives

$$\delta = 0.0317221200362845\dots,$$

so the exponent is already $1.0317221200362845\dots$. The further improvement comes from changing the splitting pattern of primes near the Golod–Shafarevich budget boundary.

A selected prime that splits in Q costs two budget units; a ramified or non-split selected prime costs one. Thus, if a small modification of T turns valuable selected split primes into non-split primes, it frees budget that can be spent on additional weight-one primes. Replacing $q \in T$ by $q^* \notin T$ changes the quadratic character of an unramified prime p by the factor

$$\left(\frac{q^*}{p}\right) \left(\frac{q}{p}\right).$$

The heuristic implemented in the repository scores a proposed swap by the transformed local scores of valuable split primes for which this factor is -1 , with a penalty

$$\frac{1}{8} \log(q^*/q)$$

for increasing the discriminant term. The search keeps the residue class modulo 4 fixed, so the parity condition is preserved automatically.

The accepted two-step move is

$$337 \mapsto 601, \quad 197 \mapsto 433.$$

It reduces the split count among the selected primes from 254 to 240, while increasing $\#S_{\mathbb{Q}}$ from 1264 to 1278. The discriminant penalty is modest:

$$\frac{1}{8} \log \left(\frac{433 \cdot 601}{197 \cdot 337} \right) = 0.146 \dots$$

The gain in usable budget more than compensates for this penalty.

The command-line interface includes a **swap-search** subcommand for this heuristic-guided search. The certificate itself remains the main mathematical object: it can be verified without trusting the heuristic that found it.

5. LIMITATIONS

This improvement is deliberately modest in scope. It is a parameter-only optimisation inside Sawin's proof, not a new proof of the unit-distance theorem. It inherits all structural features of Lemma 12 and Proposition 10 in [2], including the use of the explicit relative class-number bound of Louboutin [14], Golod–Shafarevich tower input [9, 10, 11, 15], and the prescribed local splitting conditions.

There are several concrete limitations.

- (i) The swap search is heuristic. The pair of swaps above is a robust local improvement under the implemented single-swap search, and it was also checked against a targeted double-swap exploration, but this is not an exhaustive proof of optimality over all 81-element choices of T .
- (ii) A larger T increases the Golod–Shafarevich budget, but also increases the discriminant term

$$\log \left(4 \prod_{q \in T} q \right),$$

which is penalised in the numerator. The numerical problem is therefore a balance, not a monotone gain from adding more ramified primes.

- (iii) Non-prefix choices of T make the search combinatorial. A more principled next step would be to encode simultaneous desired flips of Legendre symbols as linear constraints over $\mathbb{F}_2$, then search low-discriminant cosets. This could reveal correlated triple, quintuple or sextuple swaps that a greedy method misses.
- (iv) The verifier is numerical in the logarithmic part. The primality, congruence, ramification and budget checks are exact integer computations, but the displayed value of δ is floating-point. A formal version should replace this by interval arithmetic, with a rational lower rounding such as 0.03175.
- (v) The constant C is not optimised. As in the original theorem, the result is asymptotic along arbitrarily large values of n ; it should not be read as a useful finite- n construction.

- (vi) The method remains far below the known upper exponent $4/3$. Sawin discusses a broader ceiling for this style of argument, noting that even after setting aside some losses, one obtains a general obstruction below $4/3$ unless substantially stronger information about relative class numbers and small split primes is used. Possible improvements would likely require analytic or algebraic number theory beyond the finite parameter optimisation here, for example sharper relative class-number estimates or variants related to the ideas of Ellenberg–Venkatesh [13].

In short, the certificate shows that Sawin’s explicit exponent can be pushed from 1.014114 to 1.03175 by optimising the visible parameters and engineering the splitting pattern of T . It should be viewed as a reproducible numerical sharpening, not as evidence that the present proof architecture is close to its natural limit.

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